How-To Library

Informatica

Creating Column Profiles on Avro and Parquet Data Sources in Informatica Developer

© Copyright Informatica LLC 2014, 2021. Informatica, the Informatica logo, and Big Data Management are trademarks or registered trademarks of Informatica LLC in the United States and many jurisdictions throughout the world. A current list of Informatica trademarks is available on the web at https://www.informatica.com/trademarks.

Abstract

You can discover data on Hadoop by creating and running profiles on the data in Informatica Developer. Running a profile on any data source in the enterprise gives you a good understanding of the strengths and weaknesses of its data and metadata. A profile determines the characteristics of columns in a data source, such as value frequencies, unique values, null values, patterns, and statistics. This article describes how to create column profiles for Avro and Parquet data sources in the Developer tool.

Supported Versions

- Big Data Management® 10.2

Table of Contents

Avro and Parquet Data Sources Overview. 2
Complex Data Objects for Avro and Parquet Data Sources 2
Creating an HDFS Connection. 3
Creating a Complex File Data Object with an Avro or Parquet Data Source. 4
Running a Profile on Avro and Parquet Data Sources. 6

Avro and Parquet Data Sources Overview

Avro and Parquet are semi-structured data sources. You can create a complex file data object for Avro or Parquet data sources in the local system or Hadoop Distributed File System (HDFS). You can create a column profile on the data object in Informatica Developer.

Apache Avro is a data serialization system in binary or other data formats and the Avro data is in a format that might not be directly human-readable. Apache Parquet is a columnar storage format that can be processed in a Hadoop environment and uses a record shredding and assembly algorithm. Avro and Parquet files are either single level hierarchy files or multiple-level hierarchy files. Avro and Parquet files in HDFS might use compression formats.

You can create and run column profiles on Avro and Parquet files and use the profile results to identify data quality issues in the local system or HDFS. You can run a column profile on a single Avro or Parquet data source or on multiple Avro or multiple Parquet data sources in a folder. You can run column profiles on Avro or Parquet files in HDFS that use compression mechanism. You can create and run column profiles on big data that are in Avro or Parquet format.

Complex Data Objects for Avro and Parquet Data Sources

A complex file data object is a representation of a file in the Hadoop file system. Create a complex file data object to read or write semi-structured data to HDFS.

You can create a complex file data object on an Avro or Parquet file or on a folder that contains multiple Avro or multiple Parquet files.

Avro and Parquet files are either single-level hierarchy files or multiple-level hierarchy files. The single-level hierarchy files can use the native mode and Data Processor transformation for profiling. The multiple-level hierarchy files use only the Data Processor transformation for profiling.

To use native mode for profiling, set the resource format for the complex file reader to **Avro** or **Parquet**. In native mode, the Data Integration Service directly reads and writes Avro or Parquet resources to HDFS or local file system.

To use the Data Processor transformation for profiling, set the resource format for the complex file reader to **Binary**. When you use a binary complex file data object as a source, the Data Processor transformation parses the binary output of the complex file.

You can choose **Connection** or **File** as an access type when you create the complex file data object. To access files on a Hive connection or Hadoop Distributed File System (HDFS) connection, choose the access type as **Connection**. To access the files stored locally on your machine, choose the access type as **File**.

The following table shows the different combinations of access type and resource formats that you can choose to create a complex file data object on Avro and Parquet files:

Access Type	Resource Format	Actions to be Performed
File	Avro or Parquet	Use only for flat-structured Avro and Parquet data sources. After you create the data object, update the file path in the **File path** field to correspond to the data source on the machine where you have installed the Informatica services.
File	Binary	After you create the data object, update the following fields: - **File path** field. Enter the file path of the data source on the machine where you have installed the Informatica services. - **Input Format** field. Enter **com.informatica.avro.AvroToXML** for Avro data sources, and enter **com.informatica.parquet.ParquetToXML** for Parquet data sources.
Connection	Avro or Parquet	Choose an HDFS connection.
Connection	Binary	Choose an HDFS connection.

After you create the data object, you can choose Auto, DEFLATE, or Snappy as a compression mechanism for Avro files in HDFS and DEFLATE or Auto as a compression mechanism for Parquet files in HDFS. The DEFLATE compression format uses a combination of the LZ77 algorithm and Huffman coding and the Snappy compression format uses the LZ77-type compression format with a fixed, byte-oriented encoding.

By default, the **Auto** option is selected as a compression mechanism. The Auto option takes more time to process data because the Data Integration Service detects the compression format of the file based on the file extension, decompresses the file, and then reads the file.

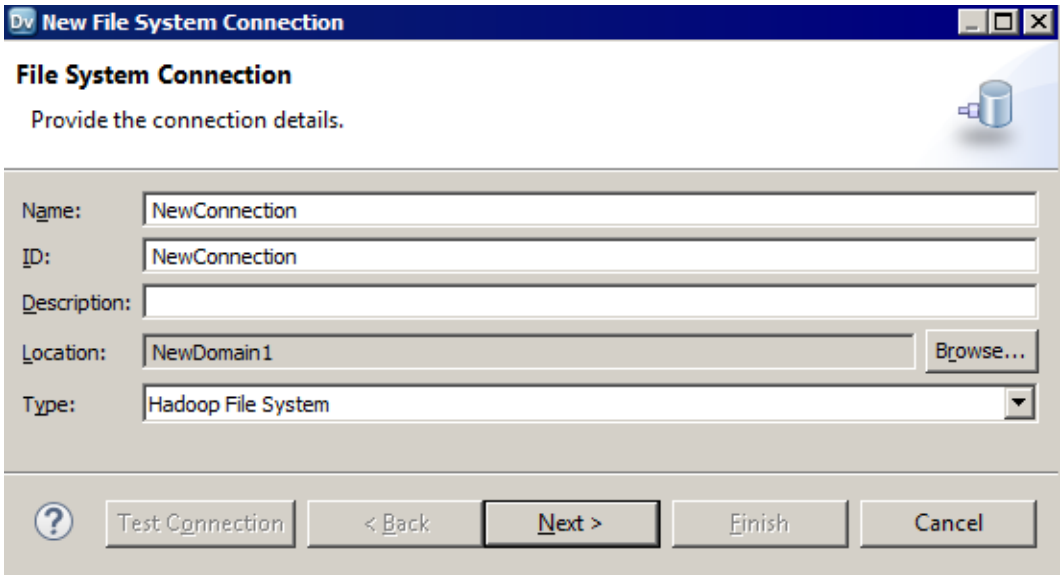
Creating an HDFS Connection

Configure the HDFS connection in Informatica Developer to create a column profile on Avro or Parquet data sources in HDFS. You can create a complex file data object after you create an HDFS connection.

1. Click **Window** > **Preferences**.
2. Select **Informatica** > **Connections**.
3. Expand the domain.
4. Select the connection type **File Systems** > **Hadoop File System**, and click **Add**.

   The **New File System Connection** dialog box appears.

   The following image shows the **New File System Connection** dialog box where you can provide the connection details to create an HDFS connection:



5. Enter a connection name.
6. Optionally, enter a connection description.
7. Click **Next**.
8. Configure the connection properties.
9. Click **Test Connection** to verify the connection to HDFS.
10. Click **Finish**.

Creating a Complex File Data Object with an Avro or Parquet Data Source

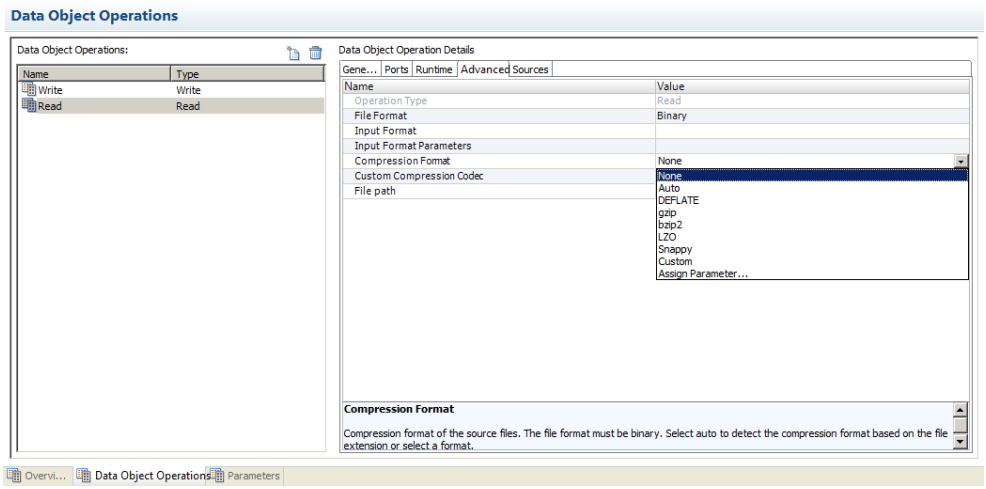
You can create a complex physical data object for an Avro or Parquet data source with **File** or **Connection** as the access type. You can create and run a column profile on the physical data object.

1. In the **Object Explorer** view, select a project.
2. Click **File** > **New** > **Data Object**.

   The **New** dialog box appears.
3. Select **Physical Data Objects** > **Complex File Data Object**, and click **Next**.

   The **New Complex File Data Object** dialog box appears.
4. Enter a name for the data object.
5. You can choose the access type as **Connection** or **File**.
   - If you choose the **Access Type** as **Connection**, then perform the following tasks:
     1. Click **Browse** to choose an HDFS connection.
        The **Choose Connection** dialog box appears.
     2. In the **Choose Connection** dialog box, choose a data source in Hive connection or HDFS connection.
        Click **OK**.
     3. In the **New Complex File Data Object** dialog box, click **Finish**.
        The data object appears in the project folder.

4. To set the compression format for the data object, select the data object in the **Projects** folder. In the **Data Object Operations** > **Read** > **Advanced** tab, select the compression format as DEFLATE or Snappy for Avro files and DEFLATE for Parquet files. By default, the **Auto** option is set.



- If you choose the **Access Type** as **File** and the **Resource Format** as **Binary**, then perform the following tasks:

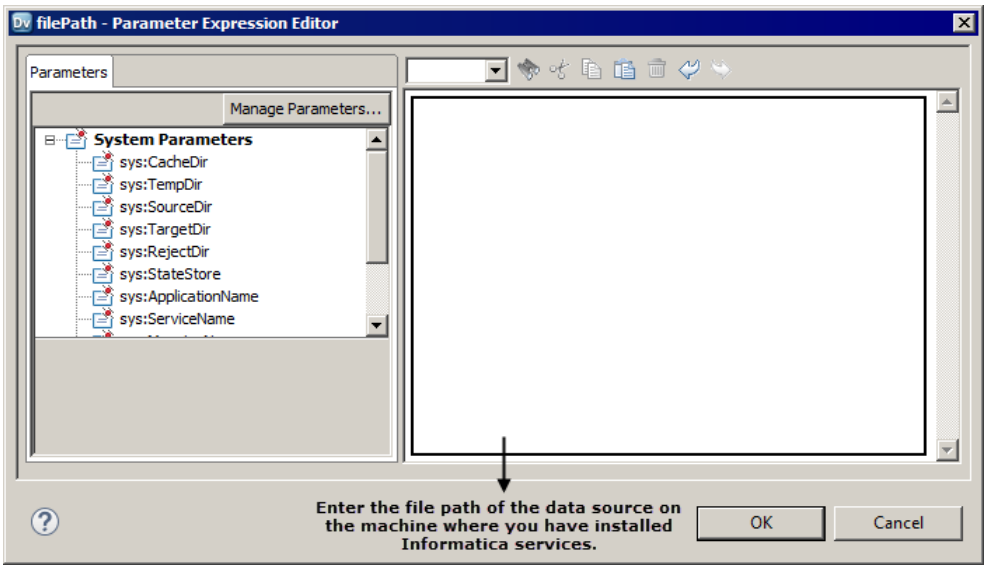
  1. Click **Browse** to choose an Avro or Parquet file on the local machine.
  2. In the **New Complex File Data Object** dialog box, click **Finish**.
     The data object appears in the project folder.
  3. Select the data object in the project folder and click the **Data Object Operations** view.
  4. In the **Data Object Operations** view, click **Read** > **Advanced** tab.
  5. In the **Advanced** tab, enter the file path of the data source on the machine where you have installed Informatica services in the **File path** field.
  6. Enter the File Format as **Custom Input**.
  7. In the **Input Format** field, enter **com.informatica.avro.AvroToXML** for Avro data sources and enter **com.informatica.parquet.ParquetToXML** for Parquet data sources. When you add the input format, the Data Processor Transformation processes and transforms the data sources in Avro or Parquet format to a data source in XML format at runtime.
     The following image shows the **Data Object Operations** view where you can edit the **Input Format** field for Avro and Parquet data sources:

Name	Value
Operation Type	Read
File Format	Custom Input
Input Format	com.informatica.avro.AvroToXML
Input Format Parameters	
Compression Format	Auto
Custom Compression Codec	

- If you choose the **Access Type** as **File** and the **Resource Format** as **Avro** or **Parquet**, then perform the following tasks:

  1. Click **Browse** to choose an Avro or Parquet file in the local machine.
  2. In the **New Complex File Data Object** dialog box, click **Finish**.
     The data object appears in the project folder.

3. After you create the data object, navigate to **Data Object Operations** > **Read** > **Advanced** tab, and verify whether the file path in the **File path** field corresponds to the data source on the machine where you have installed the Informatica services.
   The following image shows the **Parameter Expression Editor** dialog box where you can enter the file path of the data source on the machine where you have installed the Informatica services:



   Note: You can choose a folder with multiple Avro or multiple Parquet files to create a data object. After you create the data object, navigate to the **Data Object Operations** > **Read** > **Advanced** tab, and verify whether the file path in the **File path** field corresponds to the folder of the data sources on the machine where you have installed the Informatica services.

4. To view the metadata of the Avro or Parquet data source, click **Data Object Operations** > **Read** tab.

6. Click **Run** in the **Data Viewer** pane to verify whether the files is extracted and the parameters are set as expected.

   You can create a column profile on the complex file data object.

Running a Profile on Avro and Parquet Data Sources

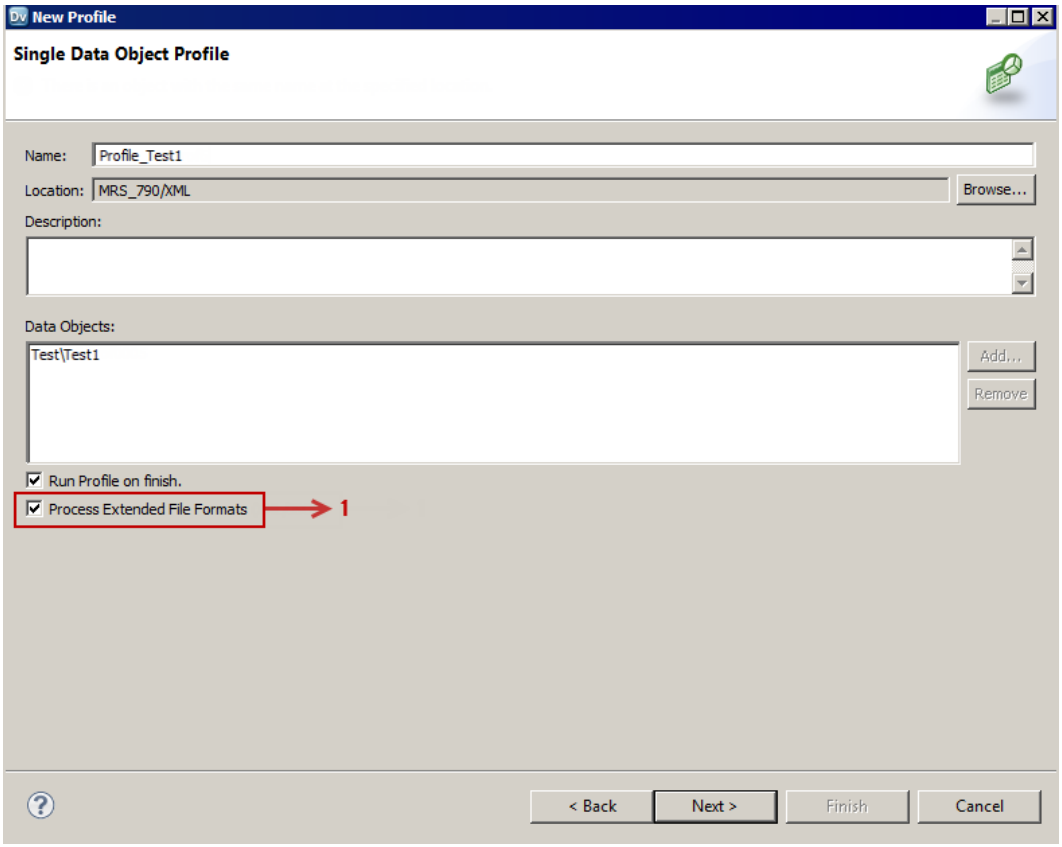
After you create a complex file data object for Avro or Parquet data sources, you can create and run a column profile on the data object.

1. In the **Object Explorer** view, select the physical data object for the Avro or Parquet data sources.
2. Click **File** > **New** > **Profile**.

   The **New** dialog box appears.
3. Select **Profile**. Click **Next**.

   The **New Profile** dialog box appears.
4. In the **New Profile** dialog box, add a name for the profile and an optional description.
5. Select **Process Extended File Formats** option. Click **Next**.

   The following image shows the **New Profile** wizard with the **Process Extended File Formats** option:



1. Process Extended File Formats. Select this option to process semi-structured data sources.

Note: The **Process Extended File Formats** option does not appear for Avro and Parquet data sources when you choose the Resource Format as **Avro** or **Parquet**.

6. In the **Single Data Object Profile** page, select the columns and options under **Column Selection** and **Data Domain Discovery** as required. Click **Finish**.

   Note: The file name of the Avro or Parquet data source appears as a column in the **Column Selection** list. If you choose the folder with multiple Avro files or multiple Parquet files to create the data object then the file names in the folder appear as column names in the **Column Selection** list. You can select the file names, if required.

7. Right-click the profile in the **Object Explorer** view, and select **Run Profile**.

   The profile results appear.
   Additionally, you can view the profile results, detect outliers, and compare profile runs in Informatica Analyst.

Author

Lavanya S